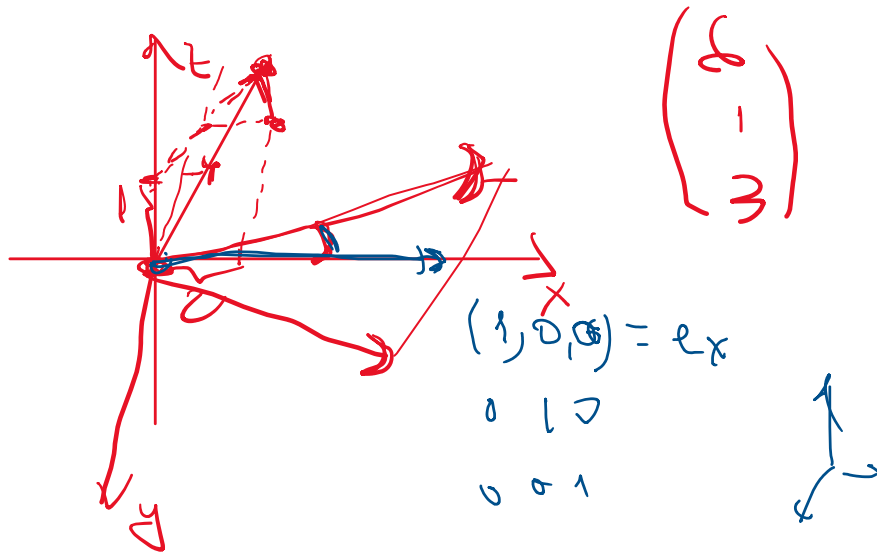


B. Exercise 1.12: Vector operations

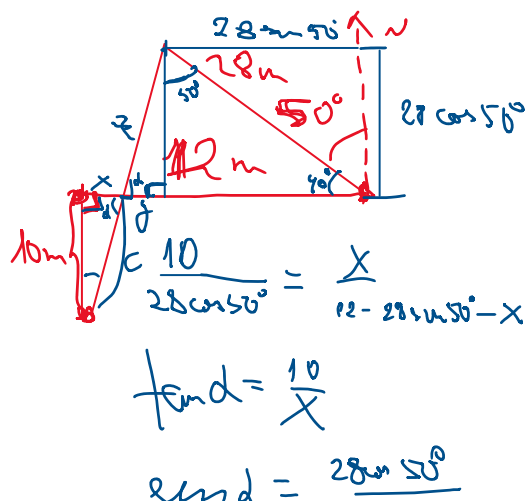
The vectors below are given without physical units. Use equal scales on the coordinate axes. In part (c), measure the angle from the positive x -axis in three dimensions. The labels c_{sum} and c_{cross} distinguish the results of parts (c) and (d).

- (a) Draw $\mathbf{u} = (2.5, 1.0)$ and $\mathbf{v} = (-0.5, -2.0)$. Plot $\mathbf{v} + \mathbf{u}$ and $\mathbf{u} - \mathbf{v}$. Calculate $\mathbf{u} \cdot \mathbf{v}$.
- (b) For $\mathbf{a} = (2, -4, 1)$ and $\mathbf{b} = (4, 5, 2)$, calculate $\mathbf{a} \cdot \mathbf{b}$.
- (c) Calculate $\mathbf{c}_{\text{sum}} = \mathbf{a} + \mathbf{b}$, its length, and its angle θ to the positive x -axis.
- (d) Calculate $\mathbf{c}_{\text{cross}} = \mathbf{a} \times \mathbf{b}$ and its length. Verify that your result is perpendicular to both inputs.



$$\cos \theta = \frac{\mathbf{c} \cdot \mathbf{e}_x}{|\mathbf{c}| |\mathbf{e}_x|} = \frac{6}{\sqrt{46}}$$

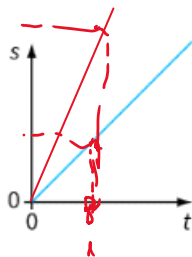
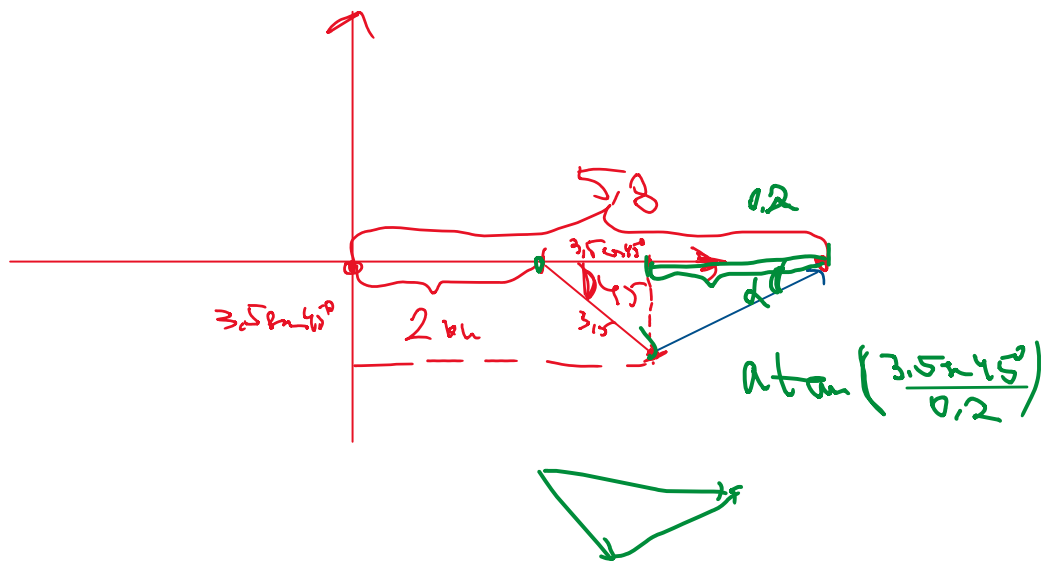
- . **A dog in an open field (Young-Freedman, Problem 1.83).** A dog runs 12.0 m east and then 28.0 m in a direction 50.0° west of north. In what direction and how far must the dog then run to end up 10.0 m south of her original starting point?



$$\sin d = \frac{28 \sin 50^\circ}{2}$$

$$\sin d = \frac{10}{2}$$

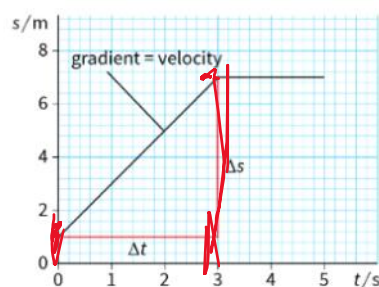
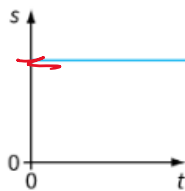
A sailboat in shifting winds (Young–Freedman, Problem 1.72). A sailor sails 2.00 km east, then 3.50 km southeast, and then an additional distance in an unknown direction. Her final position is 5.80 km directly east of the starting point. Find the magnitude and direction of the third leg of the journey. Draw the vector-addition diagram and show that it is in qualitative agreement with your numerical solution.

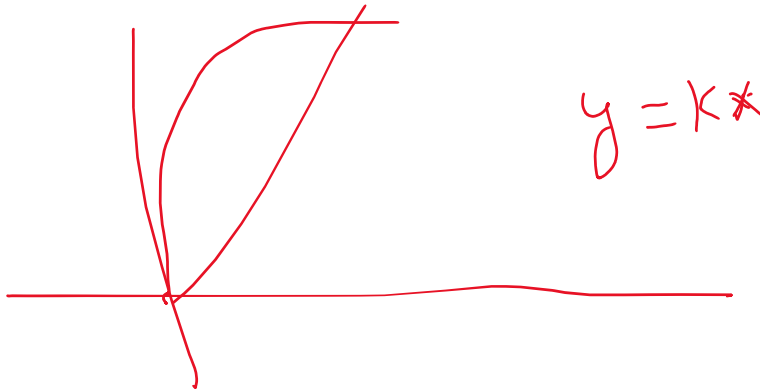
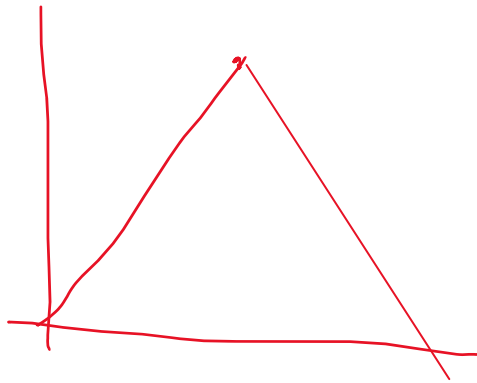


$$s = k_1 t \Rightarrow k_1 = \frac{s}{t}$$

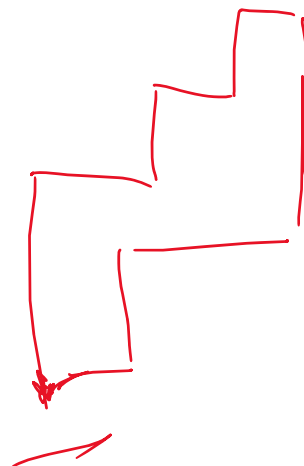
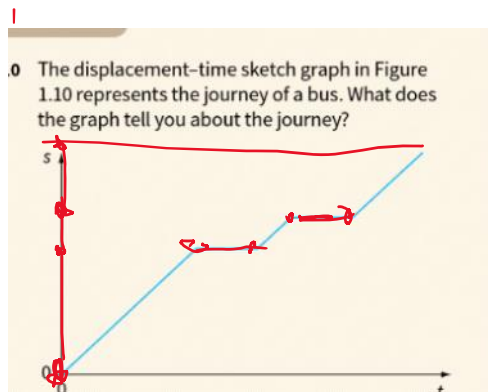
$$s = k_2 t \Rightarrow k_2 = \frac{s}{t}$$

$$k_1 < k_2$$

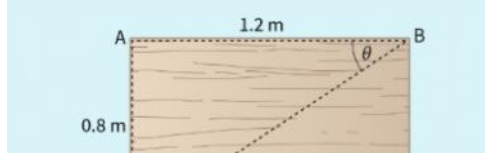




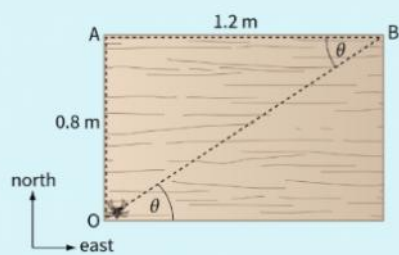
$$y = kx$$



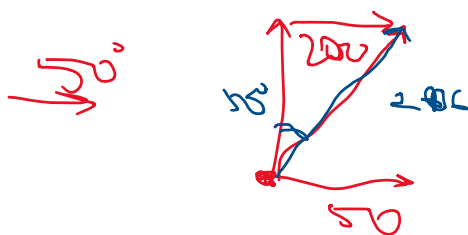
- 3 A spider runs along two sides of a table (Figure 1.13). Calculate its final displacement.



- 3 A spider runs along two sides of a table (Figure 1.13). Calculate its final displacement.



An aircraft is flying due north with a velocity of 200 m s^{-1} . A side wind of velocity 50 m s^{-1} is blowing due east. What is the aircraft's resultant velocity (give the magnitude and direction)?



$$v = \frac{\Delta s}{\Delta t} \quad a = \frac{\Delta v}{\Delta t}$$

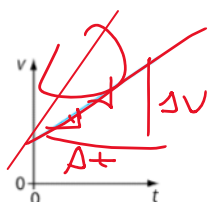
Leaving a bus stop, a bus reaches a velocity of 8.0 m s^{-1} after 10 s. Calculate the acceleration of the bus.

A sprinter starting from rest has an acceleration of 5.0 m s^{-2} during the first 2.0 s of a race. Calculate her velocity after 2.0 s.

$$a = \frac{v_f - v_i}{t}$$

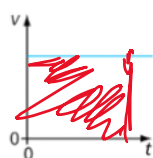
$$a t + v_i = v_f$$

A train slows down from 60 m s^{-1} to 20 m s^{-1} in 50 s. Calculate the magnitude of the deceleration of the train.

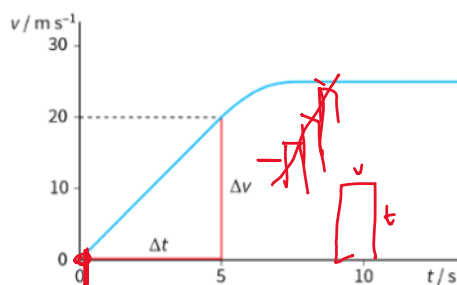
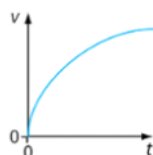


$$v = k t$$

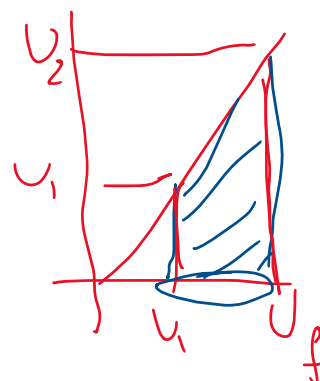
$$k = \frac{v}{t} = a$$



$v = \text{const}$
 $s = vt$



$\frac{1}{2} \Delta t \Delta v$



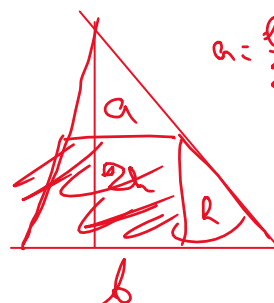
$\frac{v - u}{t} = a$

$s = \frac{1}{2} vt$

$v = u + at$
 $s = \frac{(u+v)}{2} \times t$

$v = v_f$
 $u = v_i$

$\frac{1}{2}(v_i + v_f) \cdot t$



$a = \frac{b}{2}$

$\frac{a+b}{2} \cdot h$

$\frac{b \cdot 2a}{2} = \frac{b}{2} \cdot h \cdot \frac{1}{2}$

$s = \frac{2v_i + at}{2} \cdot t$

$s = v_i t + \frac{at^2}{2}$

$v_f = v_i + a \cdot \frac{2s}{v_i + v_f}$

$v_f = v_i = \frac{2as}{v_i + v_f}$

$(a-b)(a+b) = a^2 - b^2$

$v_f^2 = v_i^2 + 2as$

$s = \frac{v_f^2 - v_i^2}{2a}$

equation 1: $v = u + at$
 $s = \frac{(u+v)}{2} \times t$

equation 1:	$v = u + at$
equation 2:	$s = \frac{(u+v)}{2} \times t$
equation 3:	$s = ut + \frac{1}{2}at^2$
equation 4:	$v^2 = u^2 + 2as$

$$g = 9.81 \text{ m/s}^2$$

To get a rough value for g , a student dropped a stone from the top of a cliff. A second student timed the stone's fall using a stopwatch. Here are their results:

estimated height of cliff = 30 m

time of fall = 2.6 s

Use the results to estimate a value for g .

$$30 = \frac{1}{2} \cdot g \cdot 2.6^2$$

18 A steel ball falls from rest through a height of 2.10 m. An electronic timer records a time of 0.67 s for the fall.

- Calculate the average acceleration of the ball as it falls.
- Suggest reasons why the answer is not exactly 9.81 m s^{-2} .

19 In an experiment to determine the acceleration due to gravity, a ball was timed electronically as it fell from rest through a height h . The times t shown in Table 2.5 were obtained.

- Plot a graph of h against t^2 .
- From the graph, determine the acceleration of free fall, g .
- Comment on your answer.

Height / m	0.70	1.03	1.25	1.60	1.99
Time / s	0.99	1.13	1.28	1.42	1.60

Table 2.5 Height (h) and time (t) data for Question 19.

- 1 A car travels one complete lap around a circular track at a constant speed of 120 km h^{-1} .
 - a If one lap takes 2.0 minutes, show that the length of the track is 4.0 km. [2]
 - b Explain why values for the **average speed** and **average velocity** are different. [1]
 - c Determine the magnitude of the displacement of the car in a time of 1.0 minute. [2]

(The circumference of a circle = $2\pi R$, where R is the radius of the circle.)

- 2 A boat leaves point A and travels in a straight line to point B (Figure 1.17). The journey takes 60 s.

Calculate:

- a the distance travelled by the boat
- b the total displacement of the boat
- c the average velocity of the boat.

Remember that each vector quantity must be given a direction as well as a magnitude.

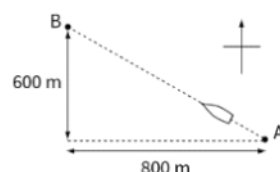


Figure 1.17 For End-of-chapter Question 2.

- 3 A boat travels at 2.0 m s^{-1} east towards a port, 2.2 km away. When the boat reaches the port, the passengers travel in a car due north for 15 minutes at 60 km h^{-1} .

Calculate:

 - a the total distance travelled [2]
 - b the total displacement [2]
 - c the total time taken [2]
 - d the average speed in m s^{-1} [2]
 - e the magnitude of the average velocity. [2]
- 4 A river flows from west to east with a constant velocity of 1.0 m s^{-1} . A boat leaves the south bank heading due north at 2.40 m s^{-1} . Find the resultant velocity of the boat. [2]
- 5 a Define **displacement**. [1]
 - b Use the definition of displacement to explain how it is possible for an athlete to run round a track yet have no displacement. [2]
- 6 A girl is riding a bicycle at a constant velocity of 3.0 m s^{-1} along a straight road. At time $t = 0$, she passes a boy sitting on a stationary bicycle. At time $t = 0$, the boy sets off to catch up with the girl. His velocity increases from time $t = 0$ until $t = 5.0 \text{ s}$, when he has covered a distance of 10 m. He then continues at a constant velocity of 4.0 m s^{-1} .
 - a Draw the displacement–time graph for the girl from $t = 0$ to 12 s. [1]
 - b On the same graph axes, draw the displacement–time graph for the boy. [2]
 - c Using your graph, determine the value of t when the boy catches up with the girl. [1]

- 7 A student drops a small black sphere alongside a vertical scale marked in centimetres. A number of flash photographs of the sphere are taken at 0.1 s intervals, as shown in Figure 1.18. The first photograph is taken with the sphere at the top at time $t = 0 \text{ s}$.
 - a Explain how Figure 1.18 shows that the sphere reaches a constant speed.
 - b Determine the constant speed reached by the sphere.
 - c Determine the distance that the sphere has fallen when $t = 0.8 \text{ s}$.

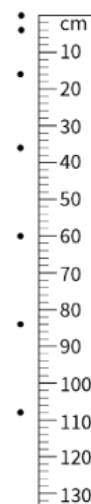


Figure 1.18 For End-of-chapter Question 7.

- 8 a State **one** difference between a scalar quantity and a vector quantity and give an example of each.
 - b A plane has an air speed of 500 km h^{-1} due north. A wind blows at 100 km h^{-1} from east to west. Draw a vector diagram to calculate the resultant velocity of the plane. Give the direction of travel of the plane with respect to north.
 - c The plane flies for 15 minutes. Calculate the displacement of the plane in this time.

- 9 A small aircraft for one person is used on a short horizontal flight. On its journey from A to B, the resultant velocity of the aircraft is 15 m s^{-1} in a direction 60° east of north and the wind velocity is 7.5 m s^{-1} due north (Figure 1.19).

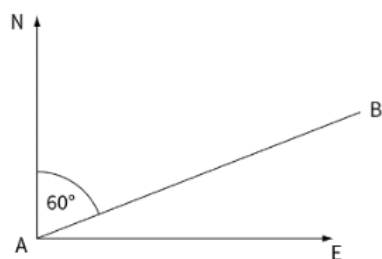


Figure 1.19 For End-of-chapter Question 9.

- Show that for the aircraft to travel from A to B it should be pointed due east.
- After flying 5 km from A to B, the aircraft returns along the same path from B to A with a resultant velocity of 13.5 m s^{-1} . Assuming that the time spent at B is negligible, calculate the average speed for the complete journey from A to B and back to A.

Thrown from the ground

A ball is thrown vertically upward from the ground with an initial speed of 20 m/s .

How long does it take to reach its maximum height?

What is its maximum height?

What is its velocity after 1.5 s ?

How long until it returns to the ground?

Finding the initial velocity

A ball is thrown vertically upward from the ground and reaches a maximum height of 45 m .

What was its initial velocity?

How long did it take to reach the maximum height?

Velocity at a given height

A stone is thrown upward from the ground at 25 m/s .

What is its velocity when it is 20 m above the ground?

There should be two possible velocities. Explain why.

4. Thrown from a building

A ball is thrown vertically upward at 15 m/s from the roof of a 30 m high building.

- Write the position function $y(t)$, taking the ground as $y = 0$.
- What is the ball's maximum height above the ground?
- How long does it take to hit the ground?
- What is its velocity just before hitting the ground?

5. Balcony problem

A student standing on a balcony 12 m above the ground throws a ball vertically upward at 10 m/s .

- What is the ball's maximum height above the balcony?
- What is its maximum height above the ground?
- How long after being thrown does it pass the balcony again?
- How long until it reaches the ground?

8. No initial velocity given

A ball is thrown upward from a 20 m high building. It reaches a maximum height of 32 m above the ground.

- Find its initial velocity.
- Find the time required to reach maximum height.

9. Find the initial height

A ball is thrown vertically upward at 12 m/s . It hits the ground 4.0 s later.

- From what height was it thrown?
- What was its maximum height above the ground?
- What was its impact velocity?

11. Given downward velocity

A ball is thrown upward from the roof of a 25 m high building. After 3.0 s , its velocity is 8.4 m/s downward.

- What was its initial velocity?
- Where is the ball at $t = 3.0 \text{ s}$?

c) Has it already passed its maximum height? How do you know?

12. Find when it reaches a window

A ball is thrown upward at 22 m/s from ground level. A window is 18 m above the ground.

a) At what time(s) does the ball pass the window?

b) What is its velocity each time?

c) Which answer corresponds to the ball moving upward and which to it moving downward?

19 In an experiment to determine the acceleration due to gravity, a ball was timed electronically as it fell from rest through a height h . The times t shown in Table 2.5 were obtained.

a Plot a graph of h against t^2 .

b From the graph, determine the acceleration of free fall, g .

c Comment on your answer.

Height / m	0.70	1.03	1.25	1.60	1.99
Time / s	0.99	1.13	1.28	1.42	1.60

Table 2.5 Height (h) and time (t) data for Question 19.